Technical Report: Grocery Sales Forecasting Architecture

Overview

This report outlines the design of our grocery sales forecasting system. Our goal was to create a reliable and accurate way to predict how many units of each product will sell across different store locations. The system is designed to act like a professional retail analyst, balancing complex mathematical models with real-world common sense.

1. How the System Works (Architecture)

We designed the system using a "modular" approach, which means we kept the data organization separate from the math used for predictions.

- **Why this matters:** This makes the system easier to maintain and improve. We can update one part of the pipeline (like how we process data) without needing to rebuild the entire system.
- **The "No-Cheating" Rule:** A common mistake in forecasting is accidentally using future information to predict the past. We designed the system with a "16-day blind spot." This forces the model to only use data that would actually be available to a store manager on the day they are making a prediction.

2. Smart Clues (Feature Strategy)

The model looks at more than just past sales numbers, it uses "clues" to understand why sales fluctuate:

- **Promotions & Foresight:** We look at how long products have been on sale and check upcoming promotional schedules. Because retailers plan these sales ahead of time, this gives the model "foresight" without breaking our "no-cheating" rule.
- **Sales Velocity:** We don’t just look at total sales; we look at the *speed* of change. We calculate if a product's popularity is accelerating or dropping, which helps us spot trends before they become obvious.
- **Real-World Context:** We include human factors like paydays (when customers typically have more money) and major events, like the 2016 earthquake, to help the model handle unusual shopping behaviour.

3. The "Two-Brain" Approach (Ensemble Philosophy)

Instead of relying on just one mathematical model, we use two different "brains" and combine their strengths:

- **The Detail-Oriented Brain (LightGBM):** This is excellent at finding specific, small patterns in large stores.
- **The Big-Picture Brain (XGBoost):** This is better at keeping predictions stable, especially for smaller or quieter stores.
- **The Result:** By combining these two (85% LightGBM and 15% XGBoost), we get a balanced prediction that is more reliable than either model could provide alone. We also perform our calculations using "log-scales," which ensures that a small neighbourhood store is treated just as fairly as a massive flagship store, so the model doesn't ignore the little guys.

Summary

I have created a reliable, "no-cheating" forecasting system that combines past data with future promotional plans and real-world context. By using a two-model approach, we ensure that our sales predictions are stable, fair, and highly accurate across all stores and product types.